



2025

L24

BOOKLET NO.

104653

Forest Services
Agriculture

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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SECTION – A

- (8×5=40)**
- Q1.** Write short notes on **any five** from seven.
- Q1.** (a) Define cropping system. Describe its types and write down the interactions between different component crops. 8
- Q1.** (b) Classify essential plant nutrients and other beneficial elements. What are the types of problem soils and their reclamation ? 8
- Q1.** (c) Describe the types and systems of farming and factors affecting them. 8
- Q1.** (d) Define agricultural extension and write down the importance and role in agriculture. 8
- Q1.** (e) Describe in detail the various methods of weed control with their examples. 8
- Q1.** (f) i) Differentiate between soil fertility and soil productivity. 8
ii) What are the principles of soil fertility ? Describe it.
- Q1.** (g) i) Define Water Use Efficiency (WUF) in relation to crop production. 8
ii) Describe in short the criteria for scheduling irrigation.
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- Q2.** (a) What is the importance of pulse crops in agriculture ? Describe the improved package of practices for higher production of Pigeonpea and Chickpea. 15
- Q2.** (b) i) Describe the modern classification of Indian soils. 15
ii) What are the minerals and organic constituents of soils ?
- Q2.** (c) Write down the scope, importance and characteristics of farm management. 10
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- Q3.** (a) i) Describe the various methods of irrigation with their advantages and disadvantages. 15
ii) Write in short climatic elements as factors on crop growth.
- Q3.** (b) What is the Lab to Land Programme (LLP) in India ? Describe it. 15
- Q3.** (c) What are the characteristics of weed ? How weeds are disseminated and multiplied ? 10
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- Q4.** (a) What are the constraints of dry farming ? Write in detail the soil moisture conservation techniques (mechanical and agronomic) in dryland agriculture. 15
- Q4.** (b) Write short on Integrated Nutrient Management (INM). Describe the judicious fertilizer use on following points. 15
i) Key aspects of judicious fertilizer use.
ii) Benefits of judicious application.
iii) Evaluation approaches.
iv) Environmental and economic consequences of injudicious use.
- Q4.** (c) Comment on marketing and pricing of agricultural inputs and outputs. 10
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SECTION – B

- Q5. Write short notes on **any five** from seven. (8×5=40)
- Q5. (a) Describe origin of genetic variation in purelines. 8
- Q5. (b) Comment on factors affecting mineral uptake by plant roots. 8
- Q5. (c) What is mulching ? Describe the purpose of mulching. 8
- Q5. (d) Describe host factors that affect the development of an epidemic. 8
- Q5. (e) Describe the steps involved in mutation breeding. 8
- Q5. (f) Describe physiological effects of ethylene. 8
- Q5. (g) What is inbreeding depression ? Write effects of inbreeding. 8
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- Q6. (a) What is heterosis ? Explain economic or useful heterosis. Explain in detail manifestation of heterosis. Differentiate between heterosis and luxuriance. 15
- Q6. (b) Describe in detail factors affecting respiration of plants. 15
- Q6. (c) State different methods of grafting and explain in detail veneer grafting and softwood grafting. 10
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- Q7. (a) Describe in detail somatic hybridization. 15
- Q7. (b) Differentiate between predators and parasites. Write families of Parasitoids with respect to their hosts. 15
- Q7. (c) What is clone ? Write characteristics of asexually propagated crops. Describe origin of genetic variation within clones. 10
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- Q8. (a) Comment on Indian Seed Act (1966). Write steps involved in release of new crop variety of field crops in Maharashtra state. 15
- Q8. (b) Describe in detail cultivation of *Amorphophallus* on following points. 15
- i) Soil and climatic requirement
 - ii) Varieties of planting
 - iii) Manures and fertilizer requirement
 - iv) Harvesting and yield.
- Q8. (c) Define systemic Fungicide. Describe in short major groups of systemic Fungicides. 10
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